

The Infrastructure phase translates your architecture design into detailed AWS resource specifications. This section focuses on creating comprehensive infrastructure specifications that can be directly implemented using Infrastructure as Code.

What You'll Learn

In this infrastructure phase, you'll:

- **Plan AWS resources** with detailed configurations and best practices
- **Design security infrastructure** with comprehensive controls and policies
- **Create deployment pipelines** with automation and quality gates
- **Implement monitoring and observability** with comprehensive metrics and alerting

Infrastructure Exercises

1. AWS Resource Planning

Create detailed specifications for all AWS resources including S3, CloudFront, API Gateway, Lambda, IAM, and Bedrock integration with exact configurations and policies.

2. Security Infrastructure

Design comprehensive security infrastructure including IAM policies, encryption, network security, and compliance controls following AWS security best practices.

3. Deployment Pipeline

Create CI/CD pipeline specifications with automated testing, deployment strategies, and environment management using AWS native tools and best practices.

4. Monitoring and Observability

Implement comprehensive monitoring, logging, and alerting strategies using CloudWatch, X-Ray, and other AWS observability services.

Key Benefits

By completing the Infrastructure phase, you'll have:

- **Detailed AWS Resource Specifications:** Complete configurations ready for deployment
- **Security-First Infrastructure:** Comprehensive security controls and compliance
- **Automated Deployment:** CI/CD pipelines with quality gates and rollback capabilities
- **Comprehensive Observability:** Monitoring, logging, and alerting for operational excellence

Advanced Features Integration

Throughout this phase, you'll leverage:

- **AWS Documentation MCP Server:** For current best practices and service configurations
- **Agent Hooks:** For validating infrastructure specifications and updating documentation
- **Agent Steering:** For applying AWS security and architectural best practices

Prerequisites

Before starting the Infrastructure exercises:

- Foundation phase completed with architecture design
- AWS account access for research and validation
- Understanding of AWS core services and pricing
- Familiarity with Infrastructure as Code concepts

Let's begin with detailed AWS resource planning that translates your architecture into specific, deployable infrastructure specifications.